Arista Access Node (BAS-ACC-62) Basavanagudi POP QoS/Congestion Event Resolution SOP

Node: BAS-ACC-62 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_2677

Description

This Standard Operating Procedure (SOP) is designed to address a Major Severity QoS/Congestion Event (Microbursts, interface tail drops, QoS queue starvation) on the Arista Access Node (BAS-ACC-62) located at the Basavanagudi POP. This failure domain is typically caused by the interaction of various protocol layers and hardware mechanisms, including but not limited to: the prioritization of traffic flows based on Differentiated Services Code Point (DSCP) values in the IPv4 and IPv6 headers, the allocation of buffer space for incoming packets, and the scheduling of packets for transmission onto the network. The Arista Access Node, being a key component of the network's edge infrastructure, plays a crucial role in policing and shaping traffic flows to prevent congestion and ensure Quality of Service (QoS) compliance. The occurrence of microbursts, interface tail drops, and QoS queue starvation can be indicative of an underlying issue with the node's configuration, hardware, or environmental conditions, such as high temperatures, power supply fluctuations, or cabling issues.

Pre-requisite Checks

- Verify that the Basavanagudi POP is accessible and that remote hands are available for assistance, if required.

- Confirm that the redundant link is healthy and functioning as expected.

- Check the power supply to the node and ensure it is receiving the correct voltage and current ratings.

- Verify that the node's temperature is within the acceptable range (typically 0°C to 40°C).

Potential Root Causes

Cause	Probability
Inadequate buffer space allocation	High
Misconfigured Quality of Service (QoS) policies	Medium
Hardware failure (e.g., faulty interface, memory leak)	Low

Diagnostic Commands & Expected Outputs

```
Command: show system resource, Expected Output: System resource utilization (e.g., CPU, memory, buffer space) to determine if any issues exist, Description: This command provides an overview of the system's resource utilization, which can help identify potential issues related to buffer space allocation or other system resources.

Command: show qos policy, Expected Output: Configured QoS policies to determine if any misconfigurations exist, Description: This command
```

```
displays the configured QoS policies, which can help identify any
misconfigurations that may be contributing to the congestion event.

Command: show interface detail, Expected Output: Interface details
(e.g., stats, errors) to determine if any interface-related issues
exist, Description: This command provides detailed information about
the interface, including statistics and error counts, which can help
identify potential hardware-related issues.
```

Resolution Steps

Step 1: Step: Adjust buffer space allocation, Command: buffer-size , Description: Increasing the buffer space allocation can help alleviate congestion by providing more space for incoming packets. However, this should be done judiciously, as excessive buffer allocation can lead to other issues, such as increased latency and packet loss.

Step 2: Step: Reconfigure QoS policies, Command: qos policy reconfigure, Description: Reconfiguring QoS policies can help ensure that traffic flows are properly prioritized and shaped, reducing the likelihood of congestion events.

Step 3: Step: Replace faulty hardware, Command: replace interface , Description: If a hardware failure is suspected, replacing the faulty interface or other hardware components can help resolve the issue.

Post-Check Verification

```
Command: show system resource, Expected Output: System resource
utilization (e.g., CPU, memory, buffer space) to verify if any issues
still exist, Description: This command is used to verify if any issues
related to buffer space allocation or other system resources still
exist after the resolution steps have been applied.

Command: show qos policy, Expected Output: Configured QoS policies to
verify if any misconfigurations still exist, Description: This command
is used to verify if any misconfigurations still exist in the QoS
policies after the resolution steps have been applied.

Command: show interface detail, Expected Output: Interface details
(e.g., stats, errors) to verify if any interface-related issues still
exist, Description: This command is used to verify if any
interface-related issues still exist after the resolution steps have
been applied.
```

Roll-back Procedures

- Step: Restore original buffer size allocation, Command: buffer-size , Description: Restoring the original buffer size allocation can help prevent further issues related to buffer space allocation.

- Step: Restore original QoS policies, Command: qos policy restore, Description: Restoring the original QoS policies can help prevent further issues related to QoS misconfigurations.

- Step: Replace original hardware, Command: replace interface , Description: Replacing the original hardware can help prevent further issues related to hardware failures.